

duced as the end product of this electrochemical reaction can be reduced. By this electrochemical reaction, the three-layered SOFC produces clean and potentially affordable power for our usage. The electricity obtained as a

result is a DC power supply, which is inverted to AC using an inverter. The heat produced can be utilised elsewhere, such as heating the building in a cold climate.

Power consumption of electrical equipment

We can estimate the average power consumption of a building from the power consumption table (on next page) and estimate the cost for installation of a bloom energy system.

A single fuel cell can produce 25W of electricity and the combination of 40 fuel cells, known as a stack, can produce up to 1kW power. The combination of these stacks makes a module which can produce power up to 25kW, and a system can produce even 100kW.

Estimate on installation costs

The current cost of each hand-made 100kW Bloom Energy server is \$700,000 to \$800,000. With mass production, the cost of each 1kW server (although 5kW would be ideal for an

As of 2010, 15 per cent of the power at eBay was created with Bloom's technology. After tax incentives that covered half the capital costs, eBay expects a three-year payback period for the remaining half, based on California's \$0.14 per kWh cost of commercial electricity.

Bloom energy is presently being used in many parts of the world. Soon we may also start using this efficient, low-cost and highly reliable energy source for our homes in India.

Cost of Bloom Box maintenance

The expected maintenance costs for the Bloom Box can be estimated based on the amount of electricity generated. The value quoted is approximately 2.4 cents per kWh, so the monthly maintenance cost is expected to be around \$22. This gives an annual maintenance cost of \$264.

Natural gas fuel cost

The current commercial Bloom Box uses 0.661 MMBtu per hour of natural gas to generate 100kW of electricity. The amount of natural gas used by a residential unit will scale linearly with the size of the unit and the amount of electricity generated. So we can estimate the amount of natural gas required by scaling linearly from the usage of 100kW. If we use bio-gas, the production cost of bloom energy can be decreased as bio-gas is cheaper than natural gas.

Operational life of Bloom Box

The Bloom Box is designed to last for 20 years. Well-maintained fuel cells can last for more than ten years. So, the expected operational life for a Bloom Box is about 10 to 20 years.

As compared to other energy sources, Bloom Energy is a good option to generate power with lower operation and maintenance charges, less air pollution and zero sound pollution. ●

Syed Zameeruddin, an independent researcher on efficient solar power generation, has patented a unique solar power converter design (patent pending). N. Pavan Kumar is an assistant professor at Sri Vishnu Engineering College for Women

Power Consumption of Electrical Equipment

Electrical equipment	Approx. consumption (watts)
Ceiling fan	60-90
Tubelight	40
Television (102cm LCD)	225
Bulb	60
Refrigerator	1000-1400
Personal computer (LCD)	120
Central air-conditioner	5000
Room air-conditioner	1000
Water heater	500
Water pump (0.5HP)	375
Iron	1100
Heavy-duty mixer grinder	600-750
Mobile phone charger	4-6
Washing machine	300-500
Microwave	500-1500

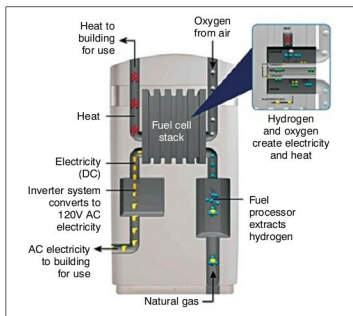


Fig. 5: A fuel cell delivering electricity to a building



Fig. 6: Power generated by a single fuel cell to a complete solution